

TECHNICAL DOCUMENTATION · REV 1.0

Autonomous Microseismic Subsidence Prediction & Early-Warning Network

A solar-powered, LoRa-mesh sensor array for continuous, unattended prediction of land subsidence in open-pit and underground mining sites, using dense low-cost vibration sensing, on-device ML noise filtering, and multi-node consensus verification.

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Reference deployment case: Jharia Coalfield, Jharkhand, India

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1. Executive summary

This document specifies an autonomous, solar-powered sensor network designed to give early warning of land subsidence at mining sites — protecting workers, equipment, and haul routes from ground failure before it becomes visible on the surface. The system works by deploying an array of vibration-sensing poles across a mine's known and moderate hazard zones. Each unit listens continuously for the earliest physical signs of underground rock instability, filters that signal locally with an on-device ML model to ignore ordinary mine noise, and cross-checks any suspected event with neighboring units before an alert is ever sent to the operator.

Because every unit is solar-powered and communicates over a long-range LoRa mesh, the array runs unattended for extended periods with no manual charging, wiring, or on-site infrastructure required. The result is a practical, scalable way to keep continuous watch over large, irregular hazard areas that would otherwise be too costly to monitor at meaningful density — giving mine operators enough lead time to evacuate personnel, halt operations, or reinforce ground ahead of a potential collapse, rather than learning about instability only after subsidence has already occurred.

Key figures at a glance

Per-unit coverage radius: **80–100 m** | Provisional unit cost: **₹1,200–1,500** | Power: fully solar/battery, zero maintenance charging | Communication: LoRa mesh | Reference site: Jharia Coalfield, full high/moderate hazard coverage estimated at **₹15–25 lakh** one-time setup.

2. Problem statement

Underground and open-pit mining removes material that was previously load-bearing, transferring stress onto the surrounding rock mass and overburden. When that stress exceeds the local strength of the rock, the ground above can subside gradually or collapse suddenly into the mined-out void (goaf). At coalfields such as Jharia, subsidence is an established and ongoing hazard: a regional hazard assessment of the 318 km² coalfield rates 1.2 km² as very-high hazard, 7.3 km² as high hazard, and 40.2 km² as moderate hazard for active subsidence.

Conventional monitoring options are poorly matched to this problem at scale. Satellite InSAR and ground-based radar cover large areas but are either slow-refresh, line-of-sight limited, or prohibitively expensive to deploy densely. Precision GNSS and geotechnical point sensors (tiltmeters, extensometers) are accurate but costly enough that only a handful of critical points can be instrumented — leaving most of a hazard zone unmonitored. What the problem actually calls for is different: a way to continuously listen for the earliest physical signs of rock failure, across a wide and irregular hazard area, cheaply enough to deploy at real density, and reliably enough to run unattended for years.

3. Sensing principle

3.1 How precursor microseismic signals are generated

Before the ground surface sinks or collapses, the underground rock mass undergoes structural failure. This process generates microseismic waves in a distinct chronological sequence:

- **Foreshock signals (weeks to days before):** as the mine roof or pillars begin to crack under stress, tiny fractures emit high-frequency microseismic waves — too small for humans to feel, but detectable by sensitive sensors.
- **Frequency shift (hours before):** shortly before a major collapse, wave frequency drops while wave energy disperses. This shift in pattern acts as a primary precursor warning of an imminent event.
- **Mainshock and collapse:** the rock layers fully fracture and drop into the mined-out void. This generates larger body and surface waves, producing measurable ground subsidence.

Physically, each micro-fracture releases a small burst of elastic strain energy that had built up in that spot — the same phenomenon as an earthquake, just enormously smaller in scale (typically magnitude -2 to -3 , corresponding to a source dimension of roughly a metre or less). This energy radiates outward as an elastic wave, travelling as a faster compressional (P) wave and a slower shear (S) wave. As mining-induced stress builds toward the point where large-scale failure becomes likely, the rock does not jump straight to collapse — it accumulates damage through many small fractures first. It is the *rate* and *spatial clustering* of these micro-events that the system's ML model is designed to track: more frequent, more tightly-clustered fracturing signals rising instability.

3.2 Why precursor signals do not travel long range

Two physical effects work against long-range detection of small events, and both are worse for small events specifically:

- **Geometric spreading:** as a wave front expands outward from its source, the same total energy is spread over an ever-larger area. Energy density falls off roughly with the square of distance. A small fracture starts with very little energy to begin with, so its signal crosses below a sensor's noise floor much sooner than a large event's would.
- **Attenuation:** rock is not a perfectly elastic medium — wave energy is continuously converted to heat as it passes through, and this loss is frequency-dependent: higher-frequency energy attenuates faster than low-frequency energy over the same distance. This matters because small, close-range fractures characteristically radiate more high-frequency energy than large events do, so the very energy that identifies a small precursor event is also the part of the signal that dies out fastest with distance.

Together, these effects mean a small fracture's signal is not just weaker at the source — it fades disproportionately fast as it travels, because the frequency band it lives in is the one attenuation removes first. Practically, this puts the reliable detection range for genuine precursor-scale micro-events at tens of metres, not hundreds — a limit set by physics, not sensor electronics.

3.3 Sensor selection: geophone vs. MEMS IMU

Two candidate sensing elements were evaluated for this system:

Feature	Analog geophone	Industrial-grade MEMS IMU (e.g. ISM330DHCX)
Primary strength	Extreme sensitivity to faint, low-frequency signals	Captures large, high-G impacts without saturating
Frequency range	Narrower — typically 4.5 Hz to 200 Hz	Ultra-wide — from 0 Hz (DC) up to 1,000+ Hz
Weakness	Clips (saturates) easily near heavy shaking	Higher noise floor — struggles with microscopic tremors
Physical setup	Sensitive to tilt; must be levelled precisely	Lightweight, cheap, functions at any installation angle

Although a geophone is well-suited to detecting very faint low-frequency vibration, its core design advantage does not match this system's target signal. Geophones are mechanically tuned to respond best at low frequency and perform best for larger, more distant events — but a close-range, small-magnitude precursor fracture radiates predominantly high-frequency energy at short range, which is exactly the regime a MEMS accelerometer is strong in. Paying the geophone premium buys sensitivity to a class of signal (large, distant, low-frequency) that is not the one this system needs to catch.

Design decision

The geophone is treated as **optional**. An industrial-grade MEMS IMU (ISM330DHCX-class) paired with an ML model trained to filter environmental noise — regular mine operations, vehicle rumble, machinery vibration — is sufficient on its own, and substantially reduces per-unit cost. Where both sensors are present on a node, their outputs are combined with **Kalman filter sensor fusion**: the geophone contributes low-frequency, high-sensitivity data while the IMU covers the wider dynamic range and prevents information loss when the geophone saturates.

3.4 Why a dense low-cost array outperforms a sparse premium array

This follows directly from the physics above, and comes down to how two variables — detection probability and event location precision — depend on sensor placement rather than sensor quality:

- **Coverage is the binding constraint, not per-node sensitivity.** Since any given micro-event is only detectable within a small radius of itself, the dominant risk is not "the sensor isn't sensitive enough" but "no sensor was close enough to this particular fracture." Adding nodes directly reduces that risk by shrinking the average distance from any random fracture point to its nearest sensor.
- **Localization needs multiple simultaneous detections anyway.** Locating a microseismic event requires the same event registering on several nearby sensors. A sparse network of excellent sensors does not help triangulation if a small event only reaches one or two of them; a dense network provides the overlapping coverage that localization structurally requires.
- **Redundancy is the anti-hallucination mechanism.** The system's council/consensus feature (Section 5.2) depends on multiple nearby nodes corroborating an event before it is trusted. This only works well with density — more overlapping neighbours means more opportunities for genuine corroboration and higher statistical confidence that a flagged event is real.
- **Budget reallocation compounds the benefit.** If a geophone-equipped node costs roughly 3–5× a MEMS-only node, the same budget buys 3–5× the number of nodes — a much better trade when coverage, not individual sensitivity, is the bottleneck.

Note: this reasoning applies specifically to a prediction system tracking trend and rate of precursor activity over time, rather than a live-event detection system requiring immediate per-event confirmation. See Section 5.1.

4. Hardware design

Each unit is built as a single field-deployable pole: a metal shaft topped with a sealed cube enclosure housing the electronics, and the sensing element embedded at the buried tip of the shaft where it sits in direct contact with the ground.

Overall pole length	1.0 m
Burial depth	0.3 m (30%) below ground — marked with a red band on the shaft as an installation guide
Exposed height	0.7 m (70%) above ground
Head enclosure	Sealed metal cube housing the microcontroller, LoRa radio, battery, and hazard LEDs
Power	Top-mounted solar panel charging an internal battery for continuous day/night operation
Sensing element	MEMS IMU (with optional geophone), embedded at the buried tip of the rod for direct ground coupling
Safety	Red hazard LEDs on two faces of the head, illuminated at night to warn mine vehicle operators of the unit's presence
Communication	LoRa radio for node-to-node mesh relay and long-range alert transmission to the operator's receiving station

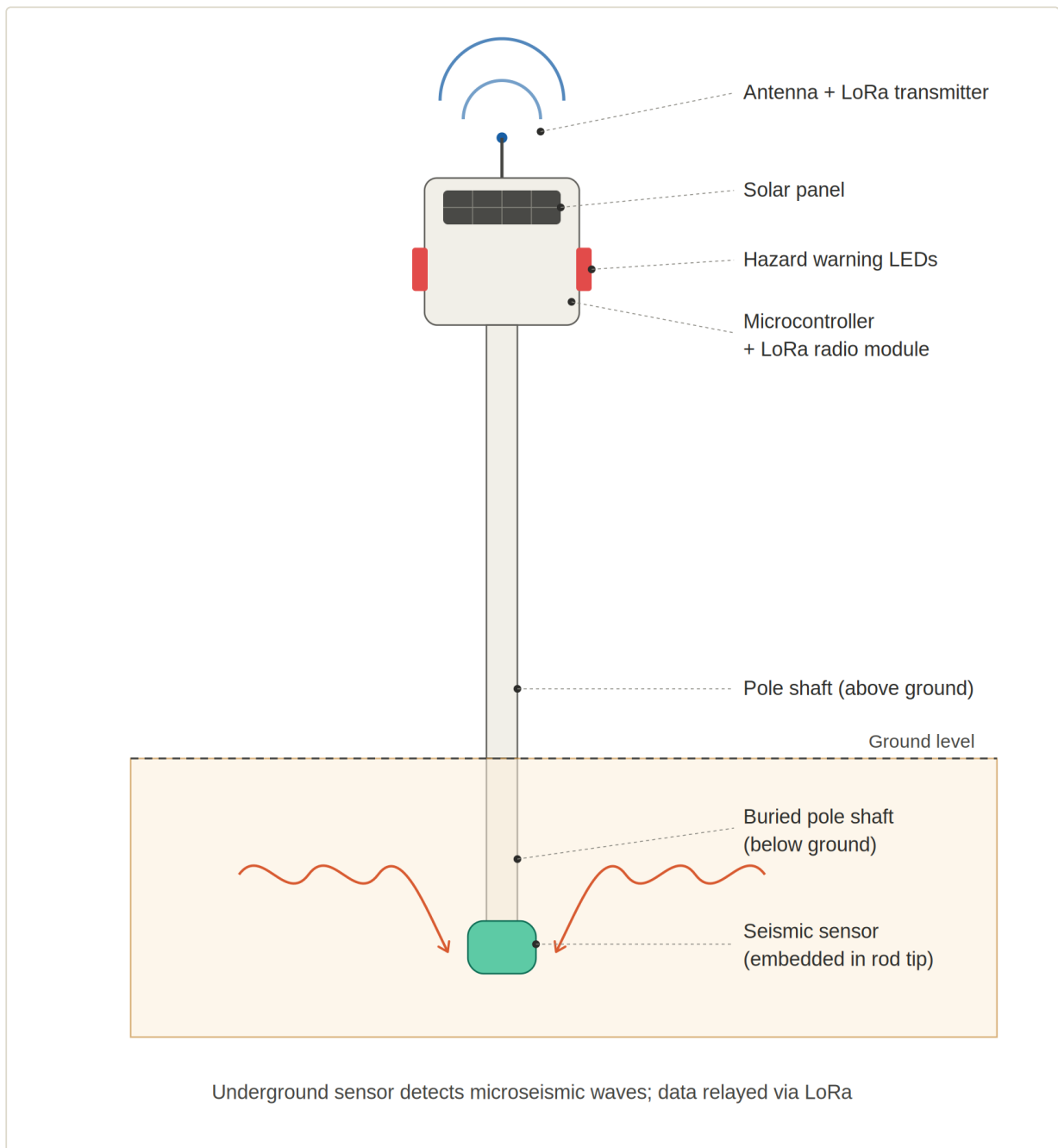


Figure 1 — Cross-section of a single unit, showing the above/below-ground split, internal components, sine-wave microseismic signal reaching the buried sensor, and the LoRa uplink from the head unit.

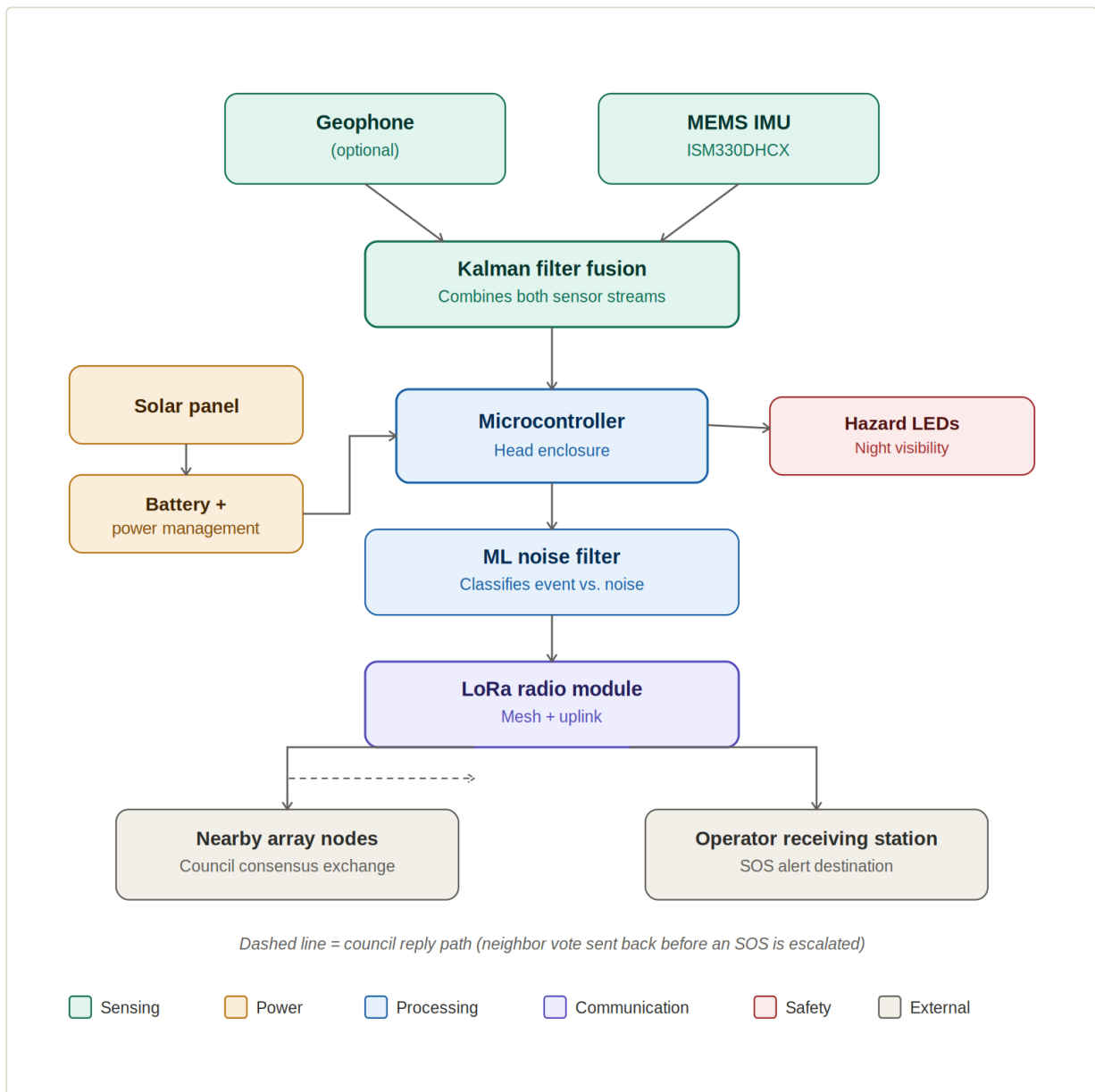


Figure 2 — Component architecture: how the sensing, power, processing, communication, and safety subsystems are wired together within a single unit, and how it links outward to nearby nodes and the operator gateway.



Figure 3 — Close-up of the deployed unit: solar-topped cube head with hazard LED strips, mounted on the ground-driven pole.

5. Detection & verification pipeline

5.1 Prediction, not live-event detection

The system's objective is to **predict and alert** ahead of a subsidence event, not to detect a live collapse as it happens. This distinction shapes the whole pipeline: rather than treating each individual vibration reading as an alert-worthy event, each node tracks statistical trends over a rolling time window — event count, cumulative energy release, spatial clustering, and rate of occurrence within its local zone. An alert is triggered when this aggregated trend crosses a threshold consistent with rising instability, not on any single reading. This relaxes real-time latency requirements considerably, which is what makes LoRa's duty-cycle limits and the council verification delay (below) acceptable for this application.

5.2 ML-based noise filtering

Raw vibration signals picked up by a node are not automatically treated as seismic events. Each suspected signal is passed through an ML model trained specifically to distinguish genuine rock-fracture microseismic activity from the everyday noise floor of an active mine — vehicle rumble, machinery operation, blasting, and other routine site activity. Only signals classified as plausible microseismic events are escalated further.

5.3 Council: multi-node consensus verification

To prevent a single faulty or "hallucinating" sensor from triggering a false alarm, no node is permitted to independently issue an SOS alert. When a unit flags a suspected positive event, it first shares that result with nearby units over LoRa. An SOS alert is escalated to the operator only once a majority of nearby devices in that region agree on the positive reading. This consensus/council mechanism is the system's primary defence against false positives, and it is precisely why array density matters (Section 3.4): more overlapping neighbours give the council more independent opinions to reach a reliable verdict from.

5.4 Sensor fusion

On units equipped with both a geophone and an IMU, the two data streams are combined using a Kalman filter. This lets the system draw on the geophone's superior sensitivity to faint, low-frequency signals while falling back on the IMU's wider dynamic range whenever the geophone saturates near a larger, closer event — giving continuous coverage across the combined sensitivity range of both sensors.

5.5 SOS decision logic

Figure 4 traces the full per-node decision path, from raw vibration signal through to a confirmed alert. Note the two independent gates a signal must clear before an operator is notified: first the local ML classification (Section 5.2), then the multi-node council vote (Section 5.3). A signal that fails either gate is logged but does not generate an alert, which is what keeps the system's false-positive rate low without requiring a human to review every candidate event.

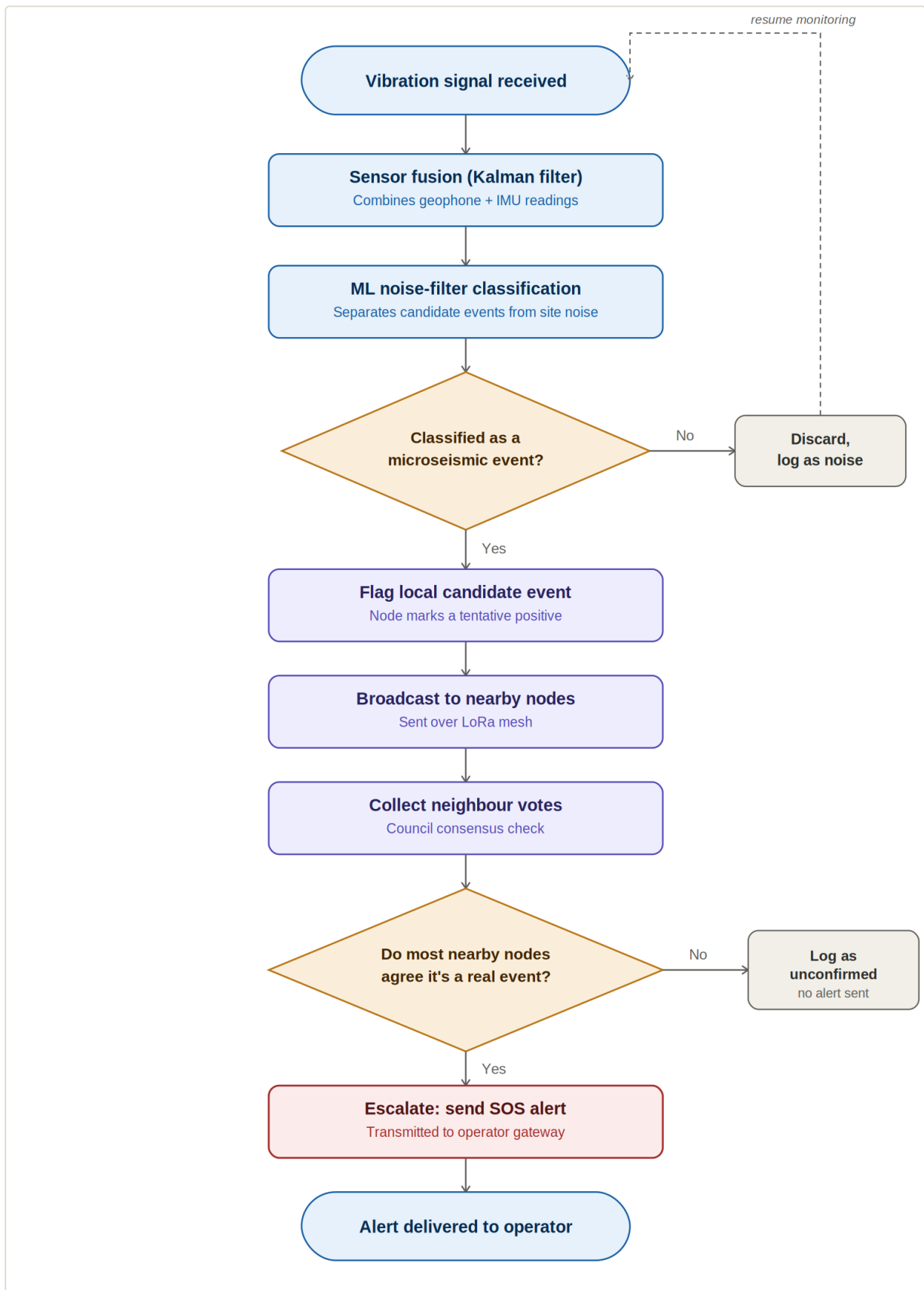


Figure 4 — Decision flow a node follows from raw vibration signal to a confirmed SOS alert, or to a logged-and-discarded non-event.

6. Network architecture

6.1 LoRa mesh communication

Units communicate using the LoRa long-range wireless protocol, both peer-to-peer (for council consensus exchanges) and back to a central receiving station at the operator's site (for SOS alert transmission). LoRa's low power draw and long effective range make it well suited to a solar-powered, unattended array, and its relatively low data rate is not a limiting factor here — nodes exchange compact event summaries (timestamp, classification, energy estimate) rather than raw waveform data, and the system's trend-based, non-real-time alerting model tolerates LoRa's duty-cycle constraints comfortably.

6.2 Array density

A single unit can realistically cover a radius of approximately 80–100 m, consistent with the tens-of-metres physical detection limit for genuine precursor-scale events established in Section 3.2, plus overlap margin for council-consensus reliability. As established in Section 3.4, a denser array of IMU-only units is the preferred configuration over a sparser array of geophone-equipped units, given the fixed budget available for any deployment.

6.3 Self-reliability

Each unit is designed for continuous, unattended operation. The top-mounted solar panel charges an internal battery during daylight hours, which then powers the unit through the night — eliminating any need for scheduled maintenance visits, manual recharging, or external power infrastructure at the deployment site.

7. Field deployment

Installation follows a fixed, repeatable procedure: the pole is driven into the ground until the red depth-marker band — set at 0.3 m from the base — reaches ground level, leaving the remaining 0.7 m, including the full head enclosure, exposed above ground. This standardised burial depth keeps the sensing tip in consistent contact with the subsurface across every unit in the array, regardless of terrain.

Because active mine sites carry continuous vehicle traffic, each unit's hazard LEDs are designed to activate automatically at night, making the low-profile pole visible to operators of mining vehicles and machinery and reducing the risk of accidental collision with a deployed unit.



Figure 5 — Array of units along a mine haul road, daytime.



Figure 6 — The same array at night; hazard LEDs illuminated for vehicle visibility.

8. Case study & cost estimation — Jharia Coalfield

The Jharia Coalfield spans approximately 318 km². Regional hazard assessment classifies subsidence risk within the field as follows:

Hazard level	Area
Very high	1.2 km ²
High	7.3 km ²
Moderate	40.2 km ²
Total targeted area	48.7 km²

At a coverage radius of 100 m per unit, fully instrumenting the very-high, high, and moderate hazard zones requires approximately **1,500–1,700 units**.

Deployment scenario	Units	Estimated one-time cost
Full coverage — very high, high & moderate zones	~1,500–1,700	₹18–25 lakh
Optimised coverage — >80% of target area	~1,200	₹15 lakh

Per-unit provisional cost is estimated at ₹1,200–1,500, based on a MEMS-IMU-only bill of materials (microcontroller, LoRa radio, IMU, solar panel, battery, enclosure) without a dedicated geophone element — consistent with the density-over-sensitivity design rationale in Section 3.4. Costs above are one-time setup estimates and do not include ongoing operational costs, which are expected to be minimal given the system's solar self-sufficiency and unattended operation.

9. Limitations & future work

- **Ground coupling dependency:** sensor accuracy depends on firm mechanical contact between the buried rod tip and surrounding soil/rock. Loose backfill around a freshly installed unit could dampen or distort incoming signals until the ground naturally settles.
- **Baseline calibration period:** because the system detects deviation from normal background microseismic activity, each deployed zone requires a period of quiet-condition data collection before trend-based predictions can be considered reliable.
- **Underground relay range:** LoRa range assumptions in this document apply to surface deployment. Any future extension to underground galleries would require closer node spacing or multi-hop relay chains, as RF propagation through rock attenuates far faster than over open ground.
- **Hybrid sensing at high-priority sites:** while the primary design favours dense MEMS-only nodes, a small number of geophone-equipped units at the highest-hazard zones remains worth evaluating for cross-validation against the wider IMU array.